



Regular Research Article

Improving Everyday Functioning in the Old-Old with Working Memory Training

Erika Borella, Ph.D., Alessandra Cantarella, Ph.D., Barbara Carretti, Ph.D.,
Annalisa De Lucia, MSc, Rossana De Beni, MSc

ARTICLE INFO

Article history:

Received November, 20 2018

Revised January, 24 2019

Accepted January, 24 2019

Key Words:

Everyday functioning
working memory
cognitive training
transfer effects
maintenance effects
old-old adults

ABSTRACT

Objective: The aim of this study was to assess gains related to working memory (WM) training, in the short and long term (9 months after the training), in abilities required in everyday life, and in cognitive measures in old-old adults (aged ≥ 75 years). **Methods:** Thirty-two community-dwelling older adults (aged 75–85 years) were randomly assigned to a training or an active control group. In addition to testing for any specific gains in a WM task similar to the one used in the training (criterion task), we sought transfer effects to: 1) abilities involved in everyday life using objective performance-based tasks (the Everyday Problem Test [EPT] and the Timed Instrumental Activities of Daily Living [TIADL] scale; 2) tasks demanding the comprehension and recall of spatial information and pairing names with faces; and 3) a measure of inhibitory control, that is, recall errors (intrusion errors). **Results:** Only the trained group showed specific gains in the criterion task, and in the TIADL in the short term. At follow-up, the trained group maintained gains in the criterion task, and showed transfer effects to everyday problem-solving (in the EPT), and in constructing spatial representations of an environment. The trained group also improved in a cognitive inhibition measure (intrusion errors) at follow-up. No such improvements were seen in the active control group. **Conclusion:** WM training may be a valid way to help old-old adults preserve at least some abilities related to everyday functioning. (Am J Geriatr Psychiatry 2019; 27:975–983)

INTRODUCTION

Working memory (WM), or the ability to maintain and simultaneously process information

for use in a complex task, is a crucial human cognitive mechanism. It is essential to everyday life functioning, but clearly undergoes a linear decline with aging.¹ Procedures to improve WM performance, along with any effects on untrained tasks (transfer

From the Department of General Psychology, University of Padova, Padua, Italy. Send correspondence and reprint requests to Erika Borella, Ph.D., Department of General Psychology, University of Padova, Via Venezia, 8, 35131, Padua, Italy. e-mail: erika.borella@unipd.it

© 2019 American Association for Geriatric Psychiatry. Published by Elsevier Inc. All rights reserved.

<https://doi.org/10.1016/j.jagp.2019.01.210>

effects), have thus been the focus of various studies conducted in the last 10 years. Although still debated, there is now a body of evidence in the aging literature to support the effectiveness of WM training procedures in producing specific gains in WM tasks similar to those used in the training,² plus some transfer effects; in the few studies that examined maintenance effects, these benefits also seemed to persist over time.³ It is noticeable, however, that only a handful of studies have examined the efficacy of WM training in those who might need it most, that is, the old-old (aged ≥ 75 years), who will account for an increasingly large proportion of the population, which is aging worldwide. Assessing the benefits of cognitive training in the old-old is also particularly important because their cognitive decline is more accentuated than in the young-old (aged 65–75 years),⁴ and they experience a gradual loss of cognitive plasticity as well.^{5,6} Studies focusing on WM training gains in the old-old alone found specific training gains, but inconsistent results in terms of transfer and maintenance effects.^{7–11} Moreover, none of these studies examined the impact of training on measures of everyday life functioning or abilities related thereto. This latter aspect has only been assessed in four studies on the young-old,^{7–10} and one covering a broad range of ages.¹¹ In some cases, self-report measures were used to examine the effect of WM training on: the perceived ability to prevent cognitive failures;^{7,11} self-reported improvements in attention in everyday contexts;¹¹ or abilities crucial to everyday life functioning, such as language or reading comprehension.^{9,10} To our knowledge, the study by Cantarella et al.⁸ is the only one to have used objective performance-based measures to assess participants' ability to solve challenging real-world problems; using the Everyday Problem Test (EPT)¹² and the Timed Instrumental Activities of Daily Living (TIADL) scale,¹³ these authors found transfer effects to the EPT. The sample included only young-old participants, however, and maintenance effects were not assessed. Given the potentially positive impact on quality of life, it seems important to examine whether WM training can sustain abilities related to everyday functioning in old-old adults, especially in everyday situations that can prove difficult for them.

The aim of the present study was consequently to assess old-old community-dwelling participants on the benefits of WM training: 1) in the WM

criterion task (specific training effect); 2) on a mechanism involved in WM, that is, cognitive inhibition, or inhibitory control (analyzing participants' ability to avoid memory errors in WM tasks); and 3) in tasks reflecting everyday life situations or crucial to everyday functioning, to identify any transfer effects.

As in Cantarella et al.,⁸ everyday functioning was examined using two objective everyday performance-based measures: the EPT,¹⁵ which demands problem-solving skills associated with daily activities; and the TIADL scale,¹⁶ which assesses the time taken to complete instrumental activities of daily living (IADL) typical of everyday life. We also tested participants on other cognitive abilities that seem important in everyday life, and are somehow related to WM functioning. We considered participants' ability to understand and memorize spatial descriptions of an environment,^{9,16} and to associate names with faces, two of the more common issues that older adults mention when asked about their memory performance.¹⁷ These two skills (environment learning and pairing faces with names) are important to day-to-day functioning, and needed in situations that we often have to manage to preserve our autonomy.

The WM training procedure proposed by Borella et al.¹⁸ was used, as it has produced consistent results in terms of both short- and long-term-specific training gains and transfer effects.³ It may also be particularly suitable for the old-old because of the limited number of sessions required. This method combines an adaptive procedure with systematic changes to the demands of the task, features that promote the engagement of different cognitive processes (encoding, maintaining and inhibiting information, simultaneously managing two tasks, sustaining and shifting attention), and that are believed to favor both learning and transfer effects by making the task constantly novel, challenging, and stimulating. We expected to see specific training benefits (improvements in the criterion task)³ and their maintenance over time, as well as transfer effects. In fact, WM is needed in everyday objective performance-based tasks,¹⁵ and in other cognitive tasks that demand processing resources similar to those used in the training. In the light of previous findings in the old-old,^{19,20} however, transfer effects might be found in the trained group not at post-test, but at follow-up.

METHODS

Participants

Thirty six (see Footnote 1) healthy adults, aged 75–85 years, who were living in the community took part in the study. They were selected on the grounds of a physical and mental health questionnaire,²¹ having scored above 9 on the Italian Multi-dimensional Assessment Checklist (SVAMA),²² which indicates a good cognitive functioning and no dementia or cognitive impairment, and they performed above the critical cutoff for their age and education in the vocabulary test²³ (Italian norms²⁴). None of the participants met the exclusion criteria suggested by Crook et al.²⁵ (i.e., a history of head trauma; any neurological or psychiatric illness; any symptomatic cardiovascular condition, breathing problems, or diseases potentially causing cognitive impairments).

Participants were randomly assigned to the trained or active control groups. Three participants in the control group dropped out (one did not complete the pre-test, and another two missed the post-test, all for health-related reasons). The groups forming the final sample¹ did not differ in terms of age, years of formal education, vocabulary test scores,^{23,24} or gender distribution (Table 1). The study was approved by the ethical board of the Padua University School of Psychology.

Materials

Specific training effect

Criterion task

The Categorization Working Memory Span (CWMS) task.²¹ This task consisted of 10 sets of word lists, each set included 20 lists of words that were divided into groups of two to six lists. Participants listened to a set of word lists audio-recorded at a rate of one word per second, and were asked to tap with their hand on the table whenever an animal noun was heard (processing phase). At the end of a set, participants recalled the last word on each list (maintenance phase). The total number of words correctly recalled was used as the measure of WM

performance. Five of the sets were used as a pre-test version, the other five at the post-test stage.

Transfer effects: cognitive inhibition

Intrusion errors.²¹

The words recalled incorrectly (i.e., erroneous recall of nontarget words), as a proportion of the number of words recalled correctly, was used as a measure of the participants' inhibitory control over no longer relevant information.⁴

Everyday objective performance-based measures

Everyday problems task (EPT).¹⁵

This task measured the ability to solve challenging real-world cognitive problems embedded in instrumental domains. Each stimulus consisted of two open-ended questions representing "real-life" situations and covering instrumental activities (i.e., household management, health, and use of medicines). Participants answered questions about how they would deal with these issues. The dependent variable was the number of correct answers given. Twenty-one stimuli/situations most closely reflecting the Italian life style were chosen from the original version; seven were used at pre-test, seven at post-test, and seven at follow-up.

TIADL scale.¹³

This task consisted of five IADL in the following domains: communication, use of money, cooking, shopping, and use of medicine. Participants were asked to simulate the actions involved in each domain (e.g., looking up a phone number) using an ad hoc kit, and time constraints were imposed (from 2–3 minutes). If participants did not complete the task within the allotted time, testing on that particular activity was terminated. The dependent variable was the time (in seconds) taken to complete the actions required for all the activities, adjusted by a time penalty calculated on the basis of a participant's accuracy in performing the tasks.¹³ A parallel version was prepared, changing the stimuli to five other activities of daily living. The two versions were distributed to participants in a counter-balanced fashion between pre- and post-test.

Cognitive tasks related to everyday situations

Comprehension and recall of spatial descriptions
(spatial descriptions task).²⁶

This task measured participants' ability to mentally construct spatial representations from a verbal description. Three equivalent texts²⁵ were used, one per session, describing outdoor environments from a route perspective (a farm, a countryside with a lake, and a nature park). Each spatial description was 10 sentences long and mentioned 10 landmarks. The experimenter read the text aloud twice, and participants were asked to recall the landmarks and their position along the path and then to answer 20 true or false questions. The dependent variable was the sum of the landmarks recalled in the right position.

Face name association task.²⁷

Participants were presented with six black and white photographs of faces, with the last name of the person depicted printed below each photograph. They were asked to study them for up to 30 seconds. Then the photographs were presented without the names, one at a time, and participants were asked to write the names previously associated with them. The dependent variable was the number of names correctly recalled and associated with the corresponding face (maximum score of six). A parallel version was developed by splitting the original task consisting of 12 items in half (six face and name pairs were used at pre-test, and six at post-test), in a counterbalanced fashion. At the follow-up, the pre-test version was used again. For all tasks except the EPT and the spatial descriptions task, the pre-test version was also used at the follow-up.

Procedure

All participants attended six individual sessions lasting a similar amount of time: the first and fifth were for the pre- and post-tests, and the sixth was the 9-month follow-up. During the other three sessions (2–4), the active control group was occupied in alternative "activities" (Table 2), while the trained group was administered the WM training. The duration of the training and the amount of interaction with the

experimenter were much the same for the two groups.

Participants in the trained group were administered a modified version of the CWMS (Table 2), in which lists of audio-recorded words were presented and participants were asked to recall target words, and also to tap with their hand on the table when they heard an animal noun. In the training sessions, the maintenance demand (the number of words that successful participants needed to recall) and the processing requirement (tapping on the table when an animal noun occurred) were manipulated (Table 2). Participants in the active control group were asked to fill in questionnaires (Table 2).

Statistical Analyses

The two groups' baseline performance was first compared using separate analyses of variance, with group (trained and active control) as the between-subjects factor, on pre-test performance in all tasks. To examine the specific training-related gains and transfer effects, univariate analysis of covariance (ANCOVA) were run using the post-test score (for immediate effects) or the follow-up score (for long-term effects) as dependent variables, the pre-test score as a covariate, and group as a between-subjects factor (see Footnote 2).

RESULTS

The results showed that the groups did not differ at the pre-test session. Descriptive statistics are shown in Table 1. The results of the ANCOVAs (Table 3) showed a main effect of group for the criterion task (specific training gain): the trained group performed better than the control group in the CWMS at both post-test and follow-up. As for transfer effects, the trained group only outperformed the control group at post-test in the TIADL, and the effect was not maintained. The trained participants then only performed better than the controls at the follow-up in the cognitive inhibition measure (with fewer intrusion errors), the EPT, and the spatial descriptions task. In the face-name association task, the main effect of group was never significant.

TABLE 1. Descriptive Pre-Test, Post-Test, and Follow-Up Data by Group

	Pre-Test						Post-Test				Follow-Up			
	Trained Group N = 18 9F			Control Group N = 14 7F			Trained versus Control Group		η^2		Trained Group		Control Group	
	M	SD		M	SD		F (1,31)				M	SD	M	SD
Age	79	3.11		79.14	3.03		<1		0.01					
Education	5.39	1.14		5.50	1.29		<1		0.02					
Vocabulary	45.44	4.87		42.07	5.10		3.61		0.10					
CW/MS task	6.06	1.89		7.57	3.52		2.44		0.07		10.44	2.33	7.36	2.47
Intrusion errors	1.31	1.75		0.85	0.81		<1		0.03		0.58	0.49	0.89	0.79
EPT	10.39	1.72		10.21	1.58		<1		0.01		11.11	1.18	10.21	2.01
TIADL test	335.33	135.90		285.76	136.82		1.04		0.03		291.30	81.94	427.81	171.41
Face and name association	2.61	1.04		2.5	0.76		<1		0.01		3.06	1.11	2.29	1.14
Comprehension and recall of spatial descriptions	2.44	1.79		1.79	1.05		1.49		0.04		2.50	1.79	1.50	1.20

Notes: F: female; M: male; SD: standard deviation.

Effect Sizes

To gain a better understanding of the extent of the benefits of training on the trained group's performance, Cohen's *d* was computed at post-test and follow-up (Table 3). For the specific effect of the training, the trained group had large effect sizes for the WM criterion task in both the immediate and the long term.

As for transfer effects, small effect sizes emerged for the cognitive inhibition measure: intrusion errors. The effect size ranged from medium in the short term to large in the long term for the EPT, whereas a large effect size emerged at post-test, and a small one at follow-up for the TIADL scale. The effect size for the spatial descriptions task was initially medium, then became large in the long term. For the face-name association task, the effect sizes were medium at post-test and small in the long term.

DISCUSSION

The aim of this study was to examine whether WM training prompted gains in WM performance, mechanisms associated with WM, and abilities related to everyday functioning (objective everyday performance-based measures), or to some cognitive mechanisms related to WM functioning (cognitive inhibition) and, in particular, tasks common in everyday life and crucial to the old-old's autonomy. To our knowledge, to date, no other studies on these issues had focused exclusively on the old-old, but it is important to sustain their cognitive functioning, and consequent autonomy in everyday life, because they are more prone to cognitive impairment, and they are expected to form an ever-larger proportion of the population.

Our results showed that participants given WM training performed better than an active control group in the criterion task at both post-test and follow-up. This confirms previous reports of specific WM training gains,^{2,3} and underscores how WM performance can improve, with a long-term maintenance effect, even in very old age. As for any transfer effects, the only immediate gain in the trained group was seen in one of the everyday performance-based measures of functional ability: the TIADL scale. No additional transfer effects were seen in the short term. Although the lack

TABLE 2. Description of the Procedure and Activities for the Trained Group and the Control Group

Type of Session And Duration	Activities for Trained Group	Activities for the Control Group
1 (pre-test; 90 minutes)	Health interview, SVAMA, vocabulary test, CWMS task, EPT, face and name association task, TIADL test, comprehension and recall of spatial descriptions.	
2 (training; 30–40 minutes)	This session was divided into three phases. In each phase, participants listened to sets of two, three, four, or five word lists (each list containing five words), and were asked to remember target words (the first or last word on each list, in serial order) and to tap their hand on the table whenever they heard an animal noun. The procedure was adaptive: participants were presented with three sets of word lists for each level of difficulty (which depended on the number of lists in a set). If they recalled the words correctly for two of the three sets, the task's difficulty was increased (up to sets of five word lists). When they were unable to do so, they moved on to the next phase of the task, which started from the easiest level (sets of two word lists). The maintenance demand of the training task differed across the three phases of this session in that participants had to recall the last word on each list during the first phase, the first word on each list during the second phase, and the last word again during the third phase.	Participants filled in the Autobiographic Memory Questionnaire (De Beni et al., 2008 ²¹)
3 (training; 30–40 minutes)	Participants listened to sets of two, three, four, or five word lists (each list containing five words), and the maintenance demand of the training task was manipulated (compared with the first training session) by asking participants to remember target words followed by a sound (which could be anywhere on the list) in the right order. Participants were presented with four sets of word lists for each level of difficulty (which depended on the number of lists in a set). Moreover, the processing requirement (i.e., tapping on the table when an animal noun occurred) was varied by changing the number of animal words in the lists. Sets of two lists could contain from two to eight animal nouns, sets of three lists could contain from four to nine animal nouns, sets of four lists could contain from six to 11 animal nouns, and sets of five lists could contain from eight to 17 animal nouns. In this session, participants were asked to complete the whole task, from the easiest to the hardest level of difficulty, regardless of their performance.	Participants filled in the Memory Sensitivity Questionnaire (De Beni et al., 2008 ²¹)
4 (training; 30–40 minutes)	Participants listened to sets of two, three, four, or five word lists (each list containing five words), and were asked to remember target words (the first or last word on each list, in serial order) and to tap their hand on the table whenever they heard an animal noun. Participants were presented with four sets of word lists for each level of difficulty (which depended on the number of lists in a set). They were asked to recall (maintenance request): 1) the last word on each list in the first set; 2) the first word on each list in the second set; 3) the last word on each list in the third set; and 4) the first word on each list in the fourth set. In this session, participants were required to complete the whole task, from the easiest to the hardest level of difficulty, regardless of their performance.	Participants filled in the Psychological Well-Being Questionnaire (BEN-SSC; De Beni et al., 2008 ²¹)
5 (post-test; 60 minutes)	CWMS, EPT, face and name association task, TIADL test, comprehension and recall of spatial descriptions.	
6 (9-month follow-up; 60 minutes)	CWMS, EPT, face and name association task, TIADL test, comprehension and recall of spatial descriptions.	

Notes: For both groups, all activities were completed within a 2-week time frame, with a fixed 2-day break between sessions. SVAMA, Italian Multidimensional Assessment Checklist; BEN-SSC, Well-being in aging questionnaire.

of any immediate transfer gains is in line with previous findings in old-old participants given WM training,^{19,20} our result for the TIADL scale seems rather intriguing—especially because the same gain was not seen in the EPT. Although the EPT and the TIADL are

related,²⁸ they tap different aspects of everyday cognition and cognitive mechanisms:²⁸ the EPT focuses on problem-solving accuracy, and mainly requires WM and reasoning abilities, whereas the TIADL has more to do with speed of execution of daily life activities,

TABLE 3. Results of One-Way ANCOVAs and Net Effect Sizes (Short-Term and Long-Term)

				Pre-Test versus Post-Test	Pre-Test versus Follow-Up		Effect Sizes	
		F ^Δ (1,32)	η _p ²		F ^Δ (1,32)	η _p ²	Short-Term	Long-Term
Specific effect								
CWMS	Covariate	5.45	0.16 ^a		7.44	0.20 ^a		
	Group	19.12	0.40 ^c	CG < TG	34.50	0.54 ^c	CG < TG	1.26 1.68
Transfer effects								
Intrusion errors	Covariate	7.76	0.21 ^b		>1	0.02		
	Group	3.66	0.11	CG = TG	7.15	0.20 ^a	CG < TG	0.46 0.28
EPT	Covariate	1.14	0.29 ^a		0.02	0.01		
	Group	2.68	0.08	CG = TG	10.80	0.27 ^b	CG < TG	0.55 1.17
TIADL	Covariate	31.92	0.52 ^c			13.35	0.31 ^c	
	Group	27.12	0.48 ^c	CG < TG ^d	2.64	0.08	CG = TG	1.03 0.28
Face and name association	Covariate	2.88	0.09		<1	0.01		
	Group	3.52	0.10	CG = TG	<1	0.02	CG = TG	0.67 0.27
Comprehension and recall of spatial descriptions	Covariate	5.05	0.16 ^a		1.80	0.06		
	Group	1.85	0.06	CG = TG	9.05	0.24 ^b	CG < TG	0.62 1.16

Notes: Cohen's d^{39} expressing the effect size of the comparisons with values corrected using the Hedges and Olkin⁴⁰ correction factor to avoid any small sample bias. CG: control group; TG, trained group.

^a $p < 0.05$.

^b $p < 0.01$.

^c $p < 0.001$.

^d Trained participants were faster than controls.

relying mainly on processing speed. Although all these cognitive mechanisms are significant predictors of everyday cognitive skills,²⁹ and affected by a decline in everyday competence with aging,³⁰ it might take longer for any change in the various complex processes (e.g., reasoning) involved in performing the EPT to become apparent in the old-old.

As for transfer effects, the immediate gain on the TIADL scale was not maintained at the follow-up, whereas a significant improvement emerged in participants' everyday problem-solving abilities (in the EPT). The different training gains seen for these two tasks at post-test or at follow-up confirm that different complex cognitive mechanisms are being tapped by these everyday performance-based measures of functional ability, as shown by the aging literature.^{14,15,28} It is noteworthy that the time penalty in the TIADL for an incomplete, wrong, or excessively slow performance was only rarely applicable, meaning that our participants probably found the activities involved quite easy, and this would explain the lack of any significant maintenance effect.

Long-term transfer effects also emerged in the measure of cognitive inhibition (intrusion errors), a mechanism related to WM functioning, and in another task (the spatial descriptions task) assessing the ability to

construct a stable mental representation of an environment, in the sense of a memory trace related to everyday functioning. These long-term effects were generally confirmed by the effect sizes calculated.

The face-name association task proved insensitive to the WM training, but performance in this task has been shown to benefit from strategy training,²⁷ so it may be that this particular ability needs to be trained using task-specific strategies, rather than a WM training that focuses on efficient processing.

Overall, the training benefits appearing in the longer term suggest that the training procedure adopted, and the manipulations applied,³ can improve cognitive inhibition, an ability also associated with a better performance in learning and planning, as well as in decision-making.^{31,32} Regarding abilities related to everyday functioning, the training seemed to foster the old-old's ability to solve accurately problems relating to daily life (e.g. to manage their medication, cope with transportation, use money, and so on)—as measured by the EPT—rather than the time to complete these activities—as in the TIADL. At the same time, improvements in the spatial description task suggest that WM training could sustain another ability crucial to everyday life, and that is the ability to navigate in an environment to reach a destination.³²

This pattern of findings extends to the old-old results previously reported for the young-old,^{8,9} but only for the long-term assessments.

That transfer effects of the training should only emerge in the longer term, consistent with other studies,³ and with reports³³ of a so-called ‘sleeper effect.’ This phenomenon, already seen in other WM training studies involving old-old participants, indicates that it can take longer for some individuals (especially among the very elderly) to learn to process information and use resources differently (more flexibly) to complete a task, and thus, show any significant improvement in their performance³³ in more complex situations, because this involves metacognitive and motivational as well as cognitive components. Nonetheless, the existence and nature of the so-called “sleeper effect” in late adulthood needs to be further explored.

Despite these intriguing and encouraging results, we have to acknowledge some limitations of the present study. The performance-based measures of functional abilities used here (EPT and TIADL) represent the best compromise between direct observation (which is more labor-intensive) and the questionnaires or self-report measures commonly used to assess autonomy in everyday life, but they are only approximations of everyday functioning, *per se*.³⁴ The requirements for multitasking and the self-initiation of behavior are less salient in performance-based measures than in real-world functioning,³⁵ and such tests are performed in an artificial setting. Nonetheless, they enable the results of training and other interventions to be assessed more objectively, avoiding the need to use strategies, and the confounding influence of familiarity and environmental supports. It would be of interest in future studies to use ecological tasks, however, or a larger number of tasks, and other IADL to assess everyday life abilities. Participants could also be asked if they perceived any real change in their everyday functioning, and how they

explained any gains, to confirm the potential of our WM training for sustaining old-old functioning. Although our sample size was in line with those of other WM training studies,³ and the analyses conducted here had been suggested and used by others,^{36,37} larger sample sizes should be recruited to replicate our findings, also adopting more classical types of analysis (i.e., repeated measures) to detect training gains. Finally, future studies should consider the role of health status (any premorbid conditions or psychiatric disorders), and its effect on perceived health for the WM training efficacy.

CONCLUSION

Our results confirm that there is room for cognitive improvement even in the so-called fourth age. Exploiting the efficacy and use of WM training could sustain a more active aging and an adequate everyday life functioning, also in terms of its potential for inducing changes in individual trajectories of cognitive decline. This might be particularly useful (although the matter remains to be explored) for individuals at risk of dementia, or with an impaired WM capacity (as in schizophrenia). Examining these issues may extend the boundaries of WM cognitive training research.

Footnote

1. An *a priori* power analysis performed with G*Power,³⁸ with an expected medium effect size and a power of 0.90, yielded 36 as the expected number of participants.
2. Given the effect size of the vocabulary test, this measure was also entered as a second covariate in the analyses, but the results did not change.

References

1. Craik FIM, Salthouse TA: *The Handbook of Aging and Cognition*. 3rd ed New York: Psychology Press, 2008
2. Karbach J, Verhaeghen P: Making working memory work: a meta-analysis of executive-control and working memory training in older adults. *Psychol Sci* 2014; 25:2027–2037
3. Borella E, Carbone E, Pastore M, et al: Working memory training for healthy older adults: the role of individual characteristics in explaining short- and long-term gains. *Front Hum Neurosci* 2017; 11:99
4. Borella E, Carretti B, De Beni R: Working memory and inhibition across the adult life-span. *Acta Psychol* 2008; 128:33–44
5. Lövdén M, Bäckman L, Lindenberger U, et al: A theoretical framework for the study of adult cognitive plasticity. *Psychol Bull* 2010; 136:659–676

6. Hertzog C, Kramer AF, Wilson RS, et al: Enrichment effects on adult cognitive development: can the functional capacity of older adults be preserved and enhanced? *Psychol Sci Public Interest* 2008; 9:1-65
7. Brehmer Y, Westerberg H, Bäckman L: Working-memory training in younger and older adults: training gains, transfer, and maintenance. *Front Hum Neurosci* 2012; 6:63
8. Cantarella A, Borella E, Carretti B, et al: Benefits in tasks related to everyday life competences after a working memory training in older adults. *Int J Geriatr Psychiatry* 2017; 32:86-93
9. Carretti B, Borella E, Zavagnin M, et al: Gains in language comprehension relating to working memory training in healthy older adults. *Int J Geriatr Psychiatry* 2013; 28:539-546
10. Payne BR, Stine-Morrow EA: The effects of home-based cognitive training on verbal working memory and language comprehension in older adulthood. *Fron Aging Neurosci* 2017; 9:256
11. Richmond LL, Morrison AB, Chein JM, et al: Working memory training and transfer in older adults. *Psychol Aging* 2011; 26:813-822
12. Willis SL, Marsiske M: *Manual for the Everyday Problems Test*. University Park, PA: Pennsylvania State University, 1993
13. Owsley C, Sloane M, McGwin JG, et al: Timed instrumental activities of daily living tasks: relationship to cognitive function and everyday performance assessments in older adults. *Gerontology* 2002; 48:254-265
14. Kimbler KJ: Everyday problem solving and instrumental activities of daily living: support for domain specificity. *Behav Sci* 2013; 3:170-191
15. Borella E, Cantarella A, Carbone E, et al: *Quotidiana-mente: la valutazione dell'autonomia funzionale e dell'auto-percezione di fallimenti cognitivi in adulti-anziani*. FrancoAngeli 2017
16. Meneghetti C, De Beni R, Pazzaglia F, et al: The role of visuo-spatial abilities in recall of spatial descriptions: a mediation model. *Learn Individ Differ* 2011; 21:719-723
17. Kaci Fairchild J, Scogin FR: Training to Enhance Adult Memory (TEAM): an investigation of the effectiveness of a memory training program with older adults. *Aging Ment Health* 2010; 14:364-373
18. Borella E, Carretti B, Riboldi F, et al: Working memory training in older adults: evidence of transfer and maintenance effects. *Psychol Aging* 2010; 25:767-778
19. Borella E, Carretti B, Zannoni G, et al: Working memory training in old age: an examination of transfer and maintenance effects. *Arch Clin Neuropsychol* 2013; 28:331-347
20. Zinke K, Zeintl M, Eschen A, et al: Potentials and limits of plasticity induced by working memory training in old-old age. *Gerontology* 2012; 58:79-87
21. De Beni R, Borella E, Carretti B, et al: BAC. Portfolio per la valutazione del benessere e delle abilità cognitive nell'età adulta e avanzata [The assessment of well-being and cognitive abilities in adulthood and aging]. Firenze, Italy: Giunti OS, 2008
22. Gallina P, Saugo M, Antoniazzi M: et al: Validazione della Scheda per la Valutazione Multidimensionale dell'Anziano (SVAMA). *Tendenze Nuove* 2006; 6:229-264
23. Wechsler D: *WAIS-R Manual: Wechsler Adult Intelligence Scale-Revised*. New York: The Psychological Corporation, 1981
24. Orsini A, Laicardi C: *WAIS-R e terza età. La natura dell'intelligenza nell'anziano: continuità e discontinuità [WAIS-R and the third age. The nature of intelligence in elderly people: continuity and discontinuity]*. Firenze, Italy: Giunti OS, 2003
25. Crook T, Bartus RT, Ferris SH, et al: Age-associated memory impairment: proposed diagnostic criteria and measures of clinical change—report of a National Institute of Mental Health work group. *Dev Neuropsychol* 1986; 2:261-276
26. Pazzaglia F, De Beni R, Meneghetti C: The effects of verbal and spatial interference in the encoding and retrieval of spatial and nonspatial texts. *Psychol Res* 2007; 71:484-494
27. Cavallini E, Pagnin A, Vecchi T: Aging and everyday memory: the beneficial effect of memory training. *Arch Gerontol Geriatr* 2003; 37:241-257
28. Borella E, Cantarella A, Joly E, et al: Performance-based everyday functional competence measures across the adult lifespan: the role of cognitive abilities. *Int Psychogeriatr* 2017; 29:2059-2069
29. Willis SL, Jay GM, Diehl M, et al: Longitudinal change and prediction of everyday task competence in the elderly. *Res Aging* 1992; 14:68-91
30. Ball K, Edwards JD, Ross LA: The impact of speed of processing training on cognitive and everyday functions. *J Gerontol B Psychol Sci Soc Sci* 2007; 62:19-31
31. De Beni R, Palladino P: Intrusion errors in working memory tasks: are they related to reading comprehension ability? *Learn Individ Differ* 2003; 12:131-143
32. Richardson JT: *Working Memory and Human Cognition, 3*. New York, NY: Oxford University Press on Demand, 1996
33. Jaeggi SM, Buschkuhl M, Shah P, et al: The role of individual differences in cognitive training and transfer. *Mem Cognit* 2014; 42:464-480
34. Moore DJ, Palmer BW, Patterson TL, et al: A review of performance-based measures of functional living skills. *J Psychiatr Res* 2007; 41:97-118
35. Glass TA: Conjugating the “tenses” of function: discordance among hypothetical, experimental, and enacted function in older adults. *Gerontologist* 1998; 38:101-112
36. Stepankova H, Lukavsky J, Buschkuhl M, et al: The malleability of working memory and visuospatial skills: a randomized controlled study in older adults. *Dev Psychol* 2014; 50:1049-1059
37. Borella E, Carretti B, Sciore R, et al: Training working memory in older adults: is there an advantage of using strategies? *Psychol Aging* 2017; 32:178-191
38. Faul F, Erdfelder E, Lang AG, et al: G*Power3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav Res Meth* 2007; 39:175-191
39. Cohen J: *Statistical Power Analysis for the Behavioral Sciences*. Hillsdale, NJ: Erlbaum, 1988
40. Hedges LV, Olkin I: *Statistical Methods for Meta-Analysis*. Orlando, FL: Academic Press, 1985